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High-Throughput Screening of Chromatographic Separations: I. Method Development and Column Modeling

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ABSTRACT: The development of purification processes for protein biopharmaceuticals is challenging due to compressed development timelines, long experimental times, and the need to survey a large parameter space. Typical methods for development of a chromatography step evaluate several dozen chromatographic column runs to optimize the conditions. An efficient batch-binding method of screening chromatographic purification conditions in a 96-well format with a robotic liquid-handling system is described and evaluated. The system dispenses slurries of chromatographic resins into filter plates, which are then equilibrated, loaded with protein, washed and eluted. This paper evaluates factors influencing the performance of this high-throughput screening technique, including the reproducibility of the aliquotted resin volume, the contact time of the solution and resin during mixing, and the volume of liquid carried over in the resin bed after centrifugal evacuation. These factors led to the optimization of a batch-binding technique utilizing either 50 or 100 μ L of resin in each well, the selection of an industrially relevant incubation time of 20 min, and the quantitation of the hold-up volume, which was as much as one quarter of the total volume added to each well. The results from the batch-binding method compared favorably to chromatographic column separation steps for a cGMP protein purification process utilizing both hydrophobic interaction and anion-exchange steps. These high-throughput screening tools can be combined with additional studies on the kinetics and thermodynamics of protein–resin interactions to provide fundamental information which is useful for defining and optimizing chromatographic separations steps.

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KEYWORDS: chromatography; method development; column modeling; high-throughput screening; batch binding

Introduction

Batch-binding experiments are a well-established methodology for optimization of chromatographic separations,

and such studies are typically used during the early stages of development to screen many different combinations of variables (Brumbaugh and Ackers, 1971; Scopes, 1987; Shepard and Tiselius, 1949). Batch binding can provide information on the thermodynamics of the product–resin interaction (including the affinity constant and binding capacity), the selectivity between the product and impurities, and binding kinetics. The simplicity of executing batch-binding experiments, the modest resin volume requirements, and the ease of running multiple test conditions in parallel enable screening a wide range of experimental space with small amounts of product. When conducted with small test tubes and 0.5–2 mL of resin, the assessment of more than several dozen different conditions by even a skilled scientist can, however, become challenging, and could require significant amounts of resin and product to perform. Using a miniaturized batch-binding configuration and robotic dispensing of liquids, a high-throughput methodology has been developed for evaluating protein binding to process chromatography resins. This strategy has expanded the application of batch-binding studies for process development, allowing for exploration of a large parameter space governing protein–resin interactions, thereby accelerating process definition.

Adsorptive purification was reported to be first used in 96-well plates in the early 1990s for solid-phase extraction (Merson, 1991; Venn et al., 2004). Since that time, adsorptive media in 96-well plates have been widely used to purify multiple products with a single purification technique, such as the purification of reactants after combinatorial chemistry (Garr et al., 2004; Pottorf and Player, 2004), and the purification of proteins and DNA from expression libraries (Albala et al., 2000; Emili and Cagney, 2000; Lesley, 2001), and affinity libraries (Li et al., 1998; Lowe 2001; Saraswat et al. 2005).

Only recently has the 96-well plate format been used to examine purification conditions by evaluating a single product with multiple purification systems. Cramer et al. have used the 96-well format extensively for screening

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displacers for displacement chromatography (Cramer et al., 2001; Liu et al., 2006; Mazza et al., 2002a; Rege et al., 2003, 2005a; Sunasara et al., 2003), but did not use 96-well filter plates, which offer significant advantages in product recovery, design flexibility, and throughput. Kramarczyk was the first to describe the application of batch binding to the 96-well filter-plate format for the development of protein purification steps, examining hydrophobic interaction chromatography (HIC) and ion exchange (IEX) resins (Kramarczyk, 2003). He also investigated the solubility of antibodies prior to a hydrophobic interaction chromatographic HTS screen. Theimann et al. also investigated the use of batch binding utilizing 96-well plates, but also did not use filter plates (Thiemann et al., 2004). Rege et al. utilized 96-well filter plates for the development of a purification process for alpha-amylase on IEX, HIC, and hydrophobic charge-induction chromatographic resins (Rege et al. 2005b). Others have described HTS applied to other systems and proteins, including monoclonal antibodies (Charlton et al., 2006; Stein and Kieseewetter, 2006). Wierling et al. have recently added the use of SELDI as an analytical tool to break the analytical bottleneck (Wierling et al., 2007). Recently, the fermentation conditions for the production of recombinant human papilloma virus were evaluated using pipet tips filled with small volumes of resin to prepare samples for analysis (Wenger et al., 2007). A multistep purification train, at 1/1000th scale, was mimicked using these devices to allow the facile evaluation of the process yield and product purity. Due to the increasing requirements of sample preparation for proteomics, such high-throughput purification devices are becoming more common (Chapman, 2005). None of these references have, however, provided a thorough, quantitative investigation of the behavior of a variety of chromatographic media in a 96-well batch-binding format.

The work described in this paper focuses on the mechanics and manipulation of a miniaturized batch-binding system for protein purification using low-pressure process-compatible resins, and the quantification of errors and non-idealities associated with the system. The stages of batch binding were conducted in a 96-well filterplate, which was configured to test conditions with as little as 50 μL of resin per well. Resin is distributed from a bulk reservoir

into the individual wells as a slurry. Subsequently, other solutions having various concentrations of protein, buffers, salts and other excipients are dispensed by the robotic liquid handler into the well containing the resin; the robotic system can deliver between 1 and 1,000 μL of solution with a precision of better than 3% and an accuracy of better than 5% (Tecan, 2008). The resin and solutions are then agitated on an integrated platform shaker for 20 min, a time at least threefold longer than typical process chromatography residence times, but significantly shorter than the duration of the load for many industrial processes. After incubation, a centrifuge evacuates the supernatant into a 96-well plate for analysis, and the damp resin may then be returned to the robotic system for another cycle of contacting, equilibration, and centrifugation. By designing these contacting cycles to mimic a chromatographic separation (with appropriate equilibration, load, wash, elution, and regeneration solutions), information relevant to the true column operation can be obtained from the equivalent of a 50–100 μL bed volume. This miniaturized system is compatible with most process chromatographic resins, allowing a wide survey of resin types during screening studies. In addition, the 96-well format is used in a broad spectrum of analytical instrumentation developed for other high-throughput applications (UV-vis spectrometry, HPLC autoinjectors, ELISAs, dynamic light scattering, etc.), which can then be used to analyze the effluents from individual cycles.

Materials and Methods

Materials

A monoclonal antibody and a Fc-fusion protein were used in these studies. The purified monoclonal antibody was a IgG1- κ with a pI of 8.3. The fusion protein was a fusion of an extracellular receptor to an IgG1 Fc. Due to extensive sialation and sulfated tyrosines, the pI of the fusion protein was between 3.0 and 3.5. The feedstream contained product-related impurities, including prospecies and hyposulfated species. The resins used in this study are shown in Table I.

Table I. Characteristics of Chromatographic Media.

Manufacturer	Resin	Backbone	Bead diameter (μm) ^a
Bio-Rad	Ceramic Hydroxyapatite, Type I	Ceramic Hydroxyapatite	40
GE Healthcare	Phenyl Sepharose 6 Fast Flow (low sub)	Agarose	45–165
	Butyl Sepharose 4 Fast Flow	Agarose	45–165
	SP Sepharose Fast Flow	Agarose	45–165
	Q Sepharose Fast Flow	Agarose	45–165
	MabSelect Protein A	Agarose	45–165
EMD Chemicals	TMAE HiCap (M)	Polymethylmethacrylate	65
Tosoh Bioscience	Butyl Toyopearl 650C	Polymethylmethacrylate	100
	Phenyl Toyopearl 650C	Polymethylmethacrylate	100

^aRanges in bead diameters are reported when available. All chromatographic media tested had polydisperse bead diameters.

Methods

The High-Throughput System

The robot used in these studies was a Tecan Genesis 150 (Tecan US, Research Triangle Park, NC). The major functionalities used include an 8-tip pipettor arm for liquid handling, an integrated UV-vis spectrophotometer, and an orbital plate shaker (TeShake, Tecan Research Triangle Park, NC). The plate shaker was set to an orbital radius of 3 mm, operated on an eccentric axis, and had a maximum speed of 1500 rpm. Typical speeds for shaking were 1350 rpm for filter plates (800 μ L with 6 mm ID) and 350–450 rpm for resin slurry vials (2.5 cm ID, #986541, Wheaton Scientific, Millville, NJ). Slower shaking rates resulted in visible settling of the resin in the wells. The disposable tips used for liquid handling have integrated conductivity probes that allow liquid level detection.

Most liquid-handling manipulations used multiple aspirations of various stock solutions into a single tip prior to dispensing. In this technique, the robotic system aspirates a stock solution, then aspirates air to leave a gap, then aspirates another solution into the same tip. The air gap prevents significant mixing between the buffers. This process can be repeated until the tip has reached its maximum volume. An air gap of 20 μ L between multiply aspirated liquids was chosen. The multiply aspirated liquids were ejected from each tip into a well in a single dispense operation. The liquid type, which determines such parameters as the aspiration and dispensing speed, was chosen to be “water,” and was found to be adequate for all liquids studied. The resin slurries were aspirated without liquid level sensing due to splashes triggering detection above the liquid level. The tips were lowered to within approximately 1 mm of the vial bottom before aspiration. For non-organic solutions and solutions without proteins, the tips were typically reused. Organic and protein solutions wetted the tips, which caused irreproducible aspiration and dispensing operations. Fresh tips were used after each aspiration and dispensing operation of organic or protein-containing solutions.

Solution contacting and supernatant collection. The filter plates used throughout the experiments were round-well 800 μ L plates with short drip-director and either of two membrane pore sizes: a 25 μ m pore-size melt blown polypropylene (Whatman, Kent, UK p/n 7700-1804) and a 0.45- μ m pore-size polypropylene membrane (Innovative Microplate, Billerica, MA, p/n F20023.) Throughout the experiments, foil adhesive tape (T592100, E&K Scientific, Santa Clara, CA) was used on the underside of the filter-plate to prevent liquid loss during shaking. This tape was removed manually prior to centrifugation. Plate sealing compatible with fully automated plate handling was not attempted in this study. Hydrophilic membranes were not used due to leakage of solution during the long equilibration times. The 25 μ m melt blown polypropylene membrane leaked when organic solvents were used. The

0.45 μ m polypropylene membrane was superior for these applications.

The liquid was evacuated from the filter-plate by centrifuging the filter-plate and collecting the supernatant in a collection plate below. The plates were evacuated by spinning for 3 min at a relative centrifugal force of 180g for the 25 μ m polypropylene plates and 1,200 g for the 0.45 μ m polypropylene plates. No mixing of individual well contents was seen with this method as determined by visual evaluation of dye solutions, unlike an alternate evacuation procedure using vacuum.

Protein Analysis

A UV-transparent microplate (#3635, Corning, Corning, NY) was used to collect the supernatant. The solution absorptivity was measured with a 96-well spectrophotometer (SpectraMax 250, Molecular Devices, Sunnyvale, CA). The path length was determined by the Pathcheck[®] feature on the SpectraMax, which uses the absorptivity of water at 998 nm to measure the pathlength. Twofold dilution series were generated within the range of 1–0.001 g/l in triplicate. The A_{320} was measured and subtracted from the A_{280} , and a standard curve was plotted. Visual inspection of these curves placed the lower linear quantitation limit in the range of 0.02–0.04 absorbance units (Wensel et al., 2008).

High-performance liquid chromatography (HPLC) assays were performed with Waters Alliance HT (2795) and Millennium 4.0 software (Waters, Milford, MA). HPLC was used to assess Fc-fusion protein concentration and purity with respect to hyposulfated isoforms and prospecies. All flow was 1.0 mL/min, and the protein was detected by UV absorbance at 214 and 280 nm.

The concentration of the Fc-fusion protein was determined by Protein A HPLC (Poros PA ImmunoDetection Sensor Cartridge, p/n 2-1001-00, equilibrated with 50 mM phosphate, 600 mM NaCl, pH 7.2; and eluted with 100 mM phosphate, 400 mM NaCl, 0.425% phosphoric acid). The levels of sulfation variants of the Fc-fusion protein were analyzed using anion-exchange HPLC (Dionex DNA Pac PA-100, 4 $\times$ 250 mm, p/n 43010, in 10 mM Tris, pH 8.10 with a 10 min NaCl gradient from 200 to 700 mM). The relative amounts of the Fc-fusion protein and its prospecies were quantified with hydrophobic interaction (HIC) HPLC (TSK Phenyl 5PW, p/n 07573, 20 mM Tris, pH 7.20, with a 40-min gradient from 1.0 to 0.0 M ammonium sulfate).

Results and Discussion

The development of a large-scale chromatographic step based on data taken from batch binding in a 96-well plate format would require characterization of many aspects of the automated screen. These aspects include the effects of mixing intensity and resin pipetting variability on resin bed

volume consistency, protein binding kinetics, and the solution hold-up after evacuation of a well. After discussion of these points, the application of the HTS system to two previously developed chromatographic steps is discussed.

Liquid Pipetting

Stock solutions of buffers, salts, and excipients allow the robotic system to vary the solution concentration in the individual wells of the plate. The robotic system was investigated to determine optimal parameters for accuracy and precision.

The precision and accuracy of liquid handling was tested for the 200 μL pipette tips by aspiration and dispensing of 195 μL of water and 5 μL of dye, and for the 1,000 μL pipette tips by 175 μL of water and 25 μL of dye. The liquid handling was tested with virgin tips as well as with tips prewetted with 200 μL of water. The resulting absorbance of the diluted samples was compared to values obtained from the same operations performed by hand pipetting with a calibrated pipette.

The order of aspiration of the solutions was important. If the dye was aspirated first, the absorbance values were more variable and lower than if the dye was aspirated last. For these studies, the solution with the lowest excipient concentration was aspirated first—typically this was water. The protein-containing solution was aspirated last. In this fashion, the lowest excipient concentration solution would be the last solution dispensed from the tip into the well, thus washing the tip of excipients and protein during the dispensing operation. This order was found to be particularly important when aspirating very small volumes of excipient or protein.

Pre-wetting tips by aspiration and dispensing of water or buffer prior to aliquotting liquids improved pipetting accuracy and precision. After prewetting the tips with water, aspiration and subsequent dispensing of water and dye using 200 μL tips showed a variation of less than 3%, or 0.15 μL , from well to well. A *t*-test showed that the robot and the manual aliquoting were not significantly different at the 95% confidence interval. Aspiration and subsequent dispensing of 175 μL of water and 25 μL of dye using 1,000 μL tips with prewetting also showed a variation of less than 3%, or 0.75 μL . Without prewetting, the dye aliquoted by the tips was 5–10% lower than that obtained by the hand pipette. Aspirating and dispensing three solutions using this multi-aspirate technique requires about 3 min for a full 96-well plate.

Resin Pipetting

The accurate and precise dispensing of resins into the 96-well filter-plate is an important aspect of using batch-binding to model chromatographic separations. The volume of resin dispensed can be affected by resin settling during the aliquotting procedure, by exclusion of the resin from

the pipette tip, by resin retention in the tip after dispensing, and by variability in the overall solution volume aliquotted.

The reproducibility of resin dispensing was first investigated for several pipette tips used on manual pipettors. The pipette tips tested included standard 200 and 1,000 μL pipette tips as well as special wide-orifice 200 μL tips (VWR, West Chester, PA). Each tip was used to pipette manually a range of Phenyl Toyopearl 650C resin slurry volumes into pre-weighed centrifuge filters (0.65 μm membrane p/n UFC30DV25, Millipore, Bedford, MA); the majority of the liquid was evacuated by centrifugation (3 min at 1,200g), and the mass of resin remaining on the filter was measured. The mass of resin was not affected by the pipette tip used, indicating consistent pipetting of the resin slurry from each tip.

All three tips performed consistently in the comparison of pipette techniques evaluated; the mass of resin dispensed increased linearly with the slurry volume pipetted ($R^2 = 0.994$, standard deviation <3%). The density of the resin slurry was determined to be 1.09 g/mL. Since the three tips performed equally, the resin slurry was assumed to be accurately dispensed by the pipette, and the slurry concentration was unaffected by the pipette tip used. The 1,000 μL pipette tip was used as a control to compare to the robotic pipetting.

The reproducibility of resin volume aliquotted by the robotic liquid-handling system was investigated with two techniques, a gravimetric method and a resin capacity method. In the first experiment, cylindrical glass vials were filled with a 50% (v/v) slurry of Phenyl Toyopearl 650C resin, and the Tecan orbital shaker was used to maintain the resin slurry in suspension at 450 rpm. The 1,000 μL disposable pipette tips (Tecan US, Research Triangle Park, NC), were submerged to within approximately 1 mm of the bottom of the shaking vials for aspiration of the resin slurry.

The resins tended to stick to the lumen of the pipette tip during the dispensing procedure, resulting in a loss of resin. This loss was mitigated on the robotic system by the aspiration of 200 μL of the packing solution prior to the aspiration of the resin slurry. Due to the air gap that the robot places between the two liquids, the packing solution and the resin slurry were not mixed in the tip. Upon dispensing the liquids, the packing solution washed the tip lumen, minimizing resin loss.

The robot dispensed 300 μL of slurry into 96 pre-weighed 2 mL centrifuge tubes (# 14231-064 VWR, West Chester, PA) to test the accuracy of slurry dispensing. The slurry volume aliquotted was calculated from the net weight of the resin in each tube utilizing the predetermined resin density of 1.09 g/mL determined from the manual pipetting described above.

The volume of the slurry dispensed to each well in a 96-well plate was consistent (Table II), where the standard deviation is less than 1% of the total slurry volume. Upon visual inspection, the level of resin did not vary appreciably between the first wells filled and the last wells filled, indicating that amount of resin was distributed consistently.

Table II. Volume of resin slurry aliquotted robotically to wells in a 96-well plate as determined gravimetrically.

	Column 1 (first wells filled) N = 8	Column 6 (middle wells filled) N = 8	Column 12 (last wells filled) N = 8	Range N = 96
Average volume	308 μ L	313 μ L	311 μ L	Max: 333 μ L
SD	0.97%	0.26%	0.03%	Min: 294 μ L

The average volume aliquotted was systematically 4% more than expected, which is in agreement with the limits set by the manufacturer of the robotic system. The range of slurry volumes dispensed spanned 294–333 μ L, or 95% to 107% of the average of the 96-well mean. The slurry volume dispensed did not depend upon the liquid level in the glass vials containing the stock resin slurry between the volume range tested, from 20 mL down to 0.6 mL (data not shown). This accuracy in resin distribution is acceptable for the application of batch-binding to modeling chromatographic separations.

The precision of dispensing a consistent volume of resin was indirectly tested by determining the amount of protein bound to each resin aliquot in a series of wells; variations in the unbound protein concentration can be attributed to inconsistent distribution of resins in each well. A 100 μ L volume of a 50% resin slurry for eight of the nine resins listed in Table I was aliquotted into a series of 12 wells of a 0.45 μ m filter-plate. The plate was centrifuged to remove the liquid and serially equilibrated three times with a moderately strong binding solution for the test Mab used in this study, according to the resin type. The antibody was then added to each well at a 5 mg/mL challenge in the appropriate solution, shaken for 20 min, and evacuated by centrifugation. The concentration of the monoclonal antibody in the supernatant was determined by UV absorbance at 280 nm, and was used to determine the amount of the antibody bound by mass balance.

The variability of protein bound is a function of the variability in the volume of resin aliquotted to each well as well as the error in pipetting the protein, which is approximately 3%. The percentage of mAb bound for each resin and the associated variability are shown in Table III.

The highest variability observed was for SP Sepharose, at 8.5%. The amount of antibody bound to this resin was also the lowest. Small absolute differences in the

supernatant concentration were likely amplified by the low degree of antibody binding, thus increasing the variability. The variability was typically less than 5%, indicating that the variability of the combination of protein and resin dispensing was less than 5%. This precision in resin distribution is acceptable for the application of batch-binding to modeling chromatographic separations.

Differences in resin bead density and diameter will affect the settling velocity of the bead, and hence the consistency of the slurry. A similar evaluation should be conducted to ensure even resin distribution before proceeding with resins that differ markedly from those tested in this study.

Binding Kinetics in Filter-Plate Format

The choice of an appropriate contact time for the batch binding will influence the accuracy of the model of large-scale column performance. The contact time should be longer than the solution residence time in a chromatographic column, yet shorter than the loading time of the column (Lacki et al., 2007). Linear velocities for many process chromatographic steps using process resins are between 100 and 300 cm/h, in columns 10–30 cm in length. Residence times for solutes are therefore between 2 and 20 min. Loading times for industrial chromatography columns can be on the order of 1–10 h. The load time depends upon the load concentration and column capacity: lower product concentrations and higher capacity resins can benefit from longer load times. Incubation times on the order of 10 h are possible with HTS systems, though molecules with shear sensitivity may be adversely affected by such long mixing times, and evaporation would be significant in uncovered plates. These long incubation times would be most useful for investigation of dilute solutions or high mass challenges.

The kinetics of protein uptake by resin in the batch-binding format were determined by comparing the amount of antibody bound under test conditions to that bound at 120 min, a time much greater than the residence time for most column operations. While even 120 min may not be sufficient in all instances to achieve true thermodynamic equilibrium, this duration is sufficiently long with respect to many process contact times that an analysis of time constants would reveal relevant non-equilibrium phenomena.

A 100 μ L volume of each 50% resin slurry was pipetted into a series of 12 wells of two filter-plates. The plates were centrifuged to remove the liquid and serially equilibrated

Table III. Well-to-well variability of resin volume based on protein bound to resin as determined by the level of protein bound.

Resin	% protein binding (average of 12 wells)	CV (% of average)
MabSelect Protein A	34	1.6
Ceramic Hydroxyapatite, Type I	56	2.4
SP Sepharose Fast Flow	11	8.5
TMAE HiCap (M)	27	3.9
Toyopearl Phenyl-650C	53	3.5
Toyopearl Butyl-650C	52	5.0
Phenyl Sepharose 6 Fast Flow (low sub)	45	2.6
Butyl Sepharose 4 Fast Flow	23	4.2

three times with strong binding solution conditions according to the resin type. The antibody was then added to a set of wells representing each resin type at total challenges of either 5 or 30 mg antibody per mL of resin. The plates were shaken at 1350 rpm for a specified time period. Antibody was added to the second set of wells, and the plates were again shaken for a set period of time. The cycle of adding protein to the remaining wells and shaking of the plates continued for 2 h for a total of 12 different incubation periods. For the final set of wells, protein was added to the wells, the plate was mixed for 10 s, and then evacuated by centrifugation 1 min after the addition of the protein. The amount of protein in each supernatant was determined by absorbance at 280 nm.

The protein uptake rates for eight resins were determined for two antibody challenges. The first challenge, 5 mg/mL of resin, is suitable for investigating protein/resin interactions at or near the linear portion of the equilibrium-binding isotherm. This resin challenge is typical of those used during early process development when material is scarce. High load challenges require significant material and time. Since selectivity typically decreases as challenge increases, resins that do not have sufficient selectivity at low load challenge can be eliminated quickly and cheaply. Resins with high selectivity are then evaluated at higher challenges.

The second challenge, at 30 mg/mL, represents typical column operating challenges for many late-stage purification processes. When the kinetics are dominated by boundary-layer mass transfer, the uptake rate in an infinite bath is an exponential decay. When intraparticle mass transfer dominates, the kinetics are more complex, and can be represented by a series of exponential decay terms. For the quantification of the incubation time in these experiments, two exponential terms appear sufficient to determine the largest time constant as follows:

$$\frac{C(t)}{C(t = 180)} = 1 - A_1 e^{-t/\tau_1} - A_2 e^{-t/\tau_2} \tag{1}$$

The low challenge condition equilibrated quickly, as shown in Figure 1A. Within 10 min, the bound antibody concentration has reached 98% of the final binding concentration determined at 120 min for each of the resins studied. The majority of the low resin challenge uptake rates were fit sufficiently with a single exponential curve, with the exception of the SP Sepharose, which required the bi-exponential to fit adequately. The remaining resins required only a single exponential fit, implying simple uptake kinetic

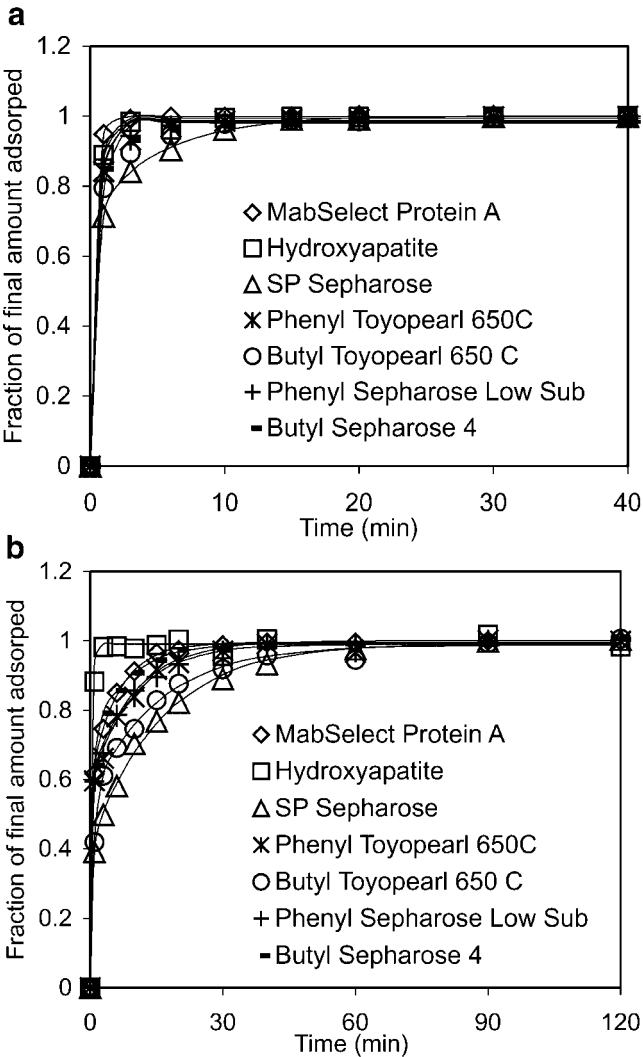


Figure 1. (a) Kinetics of antibody binding to various resins contained in a 96-well filter-plate, loaded to 5 mg of antibody per mL of resin. The timecourse continued to 120 min without change from the 20 min timepoint. The lines shown are exponential fits to the data, with the exception of the SP Sepharose, which is a bi-exponential fit. (b) Binding kinetics of the antibody to various resins contained in a 96-well filter-plate loaded to 30 mg antibody per mL of resin. The lines shown are bi-exponential fits to the data.

model. As shown in Table IVa, the longest time constant, τ_2 , in the bi-exponential fit for SP Sepharose was 5.2 min. A contact time of 20 min therefore appears to allow adequate equilibrium at 5 mg/mL challenge for all resins studied.

Table IVa. Mono- and bi-exponential fits of the uptake rate of the 5 mg/mL challenge.

Parameter	MabSelect	cHA-I	SP Sepharose	Phenyl Toyopearl 650C	Butyl Toyopearl 650C	Phenyl Sepharose High Sub	Butyl Sepharose
A_1	1.00	0.99	0.71	0.99	0.98	0.99	0.98
τ_1 (min)	0.34	0.44	0.37	0.37	0.53	0.46	0.51
A_2	NA	NA	0.29	0	0	0	0
τ_2 (min)	NA	NA	5.15	NA	NA	NA	NA

NA: The biexponential fit was not applicable to these curves.

Table IVb. Bi-exponential fit of the uptake rate of the 30 mg/mL challenge.

Parameter	MabSelect	cHA-I	SP Sepharose	Phenyl Toyopearl 650C	Butyl Toyopearl 650C	Phenyl Sepharose High Sub	Butyl Sepharose
A_1	0.59	1.00	0.41	0.60	0.55	0.58	0.71
τ_1 (min)	0.36	0.33	0.48	0.33	0.79	0.32	0.52
A_2	0.40	0.00	0.59	0.40	0.44	0.41	0.28
τ_2 (min)	6.17	10.00	16.13	10.00	15.38	9.71	8.40

Longer incubation times may be needed to approach equilibrium for higher resin challenges. Use of a 20-min contact time does not allow full equilibrium to be obtained for a 30 mg/mL challenge, as shown in Figure 1B. Again, the SP Sepharose resin has the slowest adsorption kinetics, requiring nearly 90 min to achieve 97% of the final binding. All of the resins required a bi-exponential fit, implying more complex binding kinetics than the low challenge condition. As seen in Table IVb, the longer time constant for the bi-exponential fit for SP Sepharose was 16 min. At the 20 min time point, the resins have absorbed between 82% and 100% of the amount absorbed at the final time point.

The uptake of protein may even have continued beyond the final 120 min time point. Longer equilibrium time constants for protein/resin systems have been commonly reported (Graham and Fook, 1982; Hubbuch et al., 2002; Linden et al., 1999; Yoshida et al., 1994). Under very strong binding conditions, for instance, ion exchange resins may develop a non-homogeneous distribution of bound protein (Hubbuch et al., 2002). The evaluation of equilibrium achieved by such extended incubation times is not, however, relevant to this investigation. A 20 min contact time is chosen as a conservative representation of the binding time for typical large-scale column operations.

The choice of a 20 min equilibration time may result in an underestimate of the dynamic binding capacity of the resin for the protein of interest. While capacity is a major concern in large-scale column use, operation time is also a concern. The selection of the industrially relevant incubation time of 20 min provides information on those resin/protein systems that have a useful combination of good selectivity, good capacity, and fast binding kinetics. The determination of thermodynamic information for systems with slow kinetics is not within the scope of these studies. To ensure accurate estimates of resin-protein interactions for other resins or very large proteins, a kinetic evaluation should be conducted, and an appropriate contact time determined.

Hold-Up Corrections

Quantification of the amount of liquid remaining after evacuation is important. Liquid remaining after one stage will dilute the solution added for the next stage, for instance by dilution of the load from the previous equilibration stage. Accounting for the volume of liquid carried over allows the accurate determination of the mass of protein bound to the resin and that in solution, as well as the concentration of the excipients in solution at any stage.

These quantities must be determined accurately if the data is to be used to predict column performance. Not accounting for these quantities will cause, for example, an over-estimation of the salt required for elution of a protein in ion-exchange chromatography (Rege et al., 2005b). Minimizing hold-up is also important. Large hold-up volumes cause excessive dilution of product and excipient concentrations. The use of 96-well filter plates with centrifugal evacuation rather than the standard microtiter plates used in other batch-binding studies minimizes the hold up volume. Several methods of removing the supernatant from the resin left behind in the filter-plate were examined. Vacuum filtration did not result in reproducible hold-up volumes from well to well, or from run to run. Due to the large volume of liquid left behind after removal of the supernatant by the vacuum, the solution composition cannot be changed efficiently between subsequent stages. Evacuation of supernatant via pipette from the top of the well (Cramer et al., 2001 A1; Mazza et al., 2002b) was difficult due to resin entrainment during the pipetting. Centrifugation of the supernatant through a filter-plate was determined to be both reproducible and easy.

The centrifugation method left a bed of damp resin in the well, allowing some degree of solution hold-up within the pores of the bead and in the interstices. The degree of hold-up of the solution constituents due to the remaining liquid from the previous stage can be a significant portion of the overall liquid volume.

The hold-up volume of eight resins was determined simultaneously. Three wells of a 96-well, 0.45 μm filter plate were filled with each resin, equilibrated with a high salt buffer centrifuged at 1,200g and washed with a low salt buffer. For all resins except the cHA resin, the high salt buffer consisted of 100 mM NaCl, and the low salt buffer consisted of water. For the cHA resin, the high salt buffer consisted of 8 mM Na_2HPO_4 , 100 mM NaCl, titrated to pH 7.3 and the low salt buffer consisted of 8 mM Na_2HPO_4 , 8 mM NaCl, titrated to pH 7.3. The amount of salt carried over from the equilibration into the water wash was determined by recovering the supernatant by centrifugation and measuring the salt concentration with an osmometer (Micro Osmometer Model 3300, Advanced Instruments, Norwood, MA). The use of an osmometer allowed the determination of the salt concentration in the 300 μL volume recovered from each well. A mass balance resulted in the effective hold-up volume for each, shown in Table V, normalized to a 100 μL bed volume.

In the absence of bead compression, the evacuation of the wells by centrifugation is not expected to remove the liquid

Table V. Carryover volume per 100 μL bed volume of various resins for solution components.

Resin	Small molecule carryover volume per 100 μL resin bed	
	V_{CO} (μL)	STD
MabSelect Protein A	71.2	6.4%
Ceramic Hydroxyapatite, Type I	72.0	0.3%
SP Sepharose Fast Flow	59.8	1.4%
TMAE HiCap (M)	51.8	5.2%
Toyopearl Phenyl-650C	59.8	5.8%
Toyopearl Butyl-650 C	57	3.5%
Phenyl Sepharose (Low Sub)	72.2	3.4%
Butyl Sepharose 4 Fast Flow	76.0	7.0%

from within the bead, as the centrifugal forces would not overcome the surface tension in very small pores. Assuming the bed void volume fraction to be 0.38, and the void volume of the resin beads to be between 0.70 and 0.9, the volume of liquid in the pores of a 100 μL bed of resin would be between 44 and 56 μL . The hold-up volumes reported in Table V for salt indicate that most, but not all, of the hold-up volume was due to the liquid within the bead for about half of the resins tested. The remaining resins had a significant amount of remaining interstitial liquid.

With supernatant:resin phase ratios typically ranging from 1:1 to 6:1, the hold-up volumes determined above would account for as much as 47% (a 1:1 ratio with Butyl Sepharose 4 Fast Flow) or as little as 8% (a 6:1 with TMAE) of the total liquid in the next stage. The degree of solution hold-up will also depend on certain characteristics of the chromatography resin, including pore size, porosity, and possibly compressibility, and solution characteristics such as surface tension or viscosity.

In general, the concentration of solute in an equilibrium stage, n , accounting for hold-up from the previous equilibrium stage, $n - 1$, is as follows:

$$C_{\text{solute},n} = \frac{V_{\text{solute},n} \times C_{\text{solute,stock}} + V_{\text{hold-up}} \times C_{\text{solute},n-1}}{V_{\text{hold-up}} + V_{\text{liquid}}} \quad (2)$$

where $C_{\text{solute},n}$ is the concentration of the solute desired in the current equilibrium stage, $V_{\text{solute},n}$ is the volume of the solute stock solution to be added to stage n , $C_{\text{solute,stock}}$ is the concentration of solute in the stock solution, $V_{\text{hold-up}}$ is the hold-up volume, $C_{\text{solute},n-1}$ is the solute concentration from the previous equilibrium stage, and V_{liquid} is the total liquid volume to be added to the well.

With certain exceptions, one can achieve the target concentration of solute in an equilibrium stage by increasing or decreasing the volume of stock solute solution added depending on whether the next stage is at higher or lower solute concentrations. The volume of solute stock solution in any subsequent stage can be adjusted based on the

amount of solute carried over from the previous equilibrium stage as follows:

$$V_{\text{solute},n} = \frac{(C_{\text{target},n}(V_{\text{liquid}} + V_{\text{hold-up}}) - V_{\text{hold-up}} \times C_{\text{solute},n-1})}{C_{\text{solute,stock}}} \quad (3)$$

where $C_{\text{target},n}$ is the target concentration of solute in the current equilibrium stage, n . The remaining volume needed to make up the total volume V_{liquid} is made up from a solute-free solution. One cannot achieve a target solute concentration less than

$$C_{\text{target},n} = \frac{C_{\text{solute},n-1} \times V_{\text{hold-up}}}{V_{\text{hold-up}} + V_{\text{liquid}}} \quad (4)$$

For example, when the hold-up volume is 20% of the liquid volume added, the n stage solution concentration, $C_{\text{solute},n}$, cannot fall below 16.7% of the previous stage concentration, $C_{\text{solute},n-1}$.

These relationships are invalid if the solute binds to the resin, such as in the case of imidazole in an IMAC resin. They also do not apply for pH, as the pH of the supernatant of a stage will not vary linearly with the solution concentration, as assumed above. The volumes listed in Table V are valid for non-binding small molecules. The hold-up volume for macromolecules may be smaller due to exclusion of the molecule from some small pores. Large proteins may require the hold-up volume to be measured using the target protein.

Batch-Binding Models of Chromatographic Column Separations

The robotic system was evaluated as a model for two chromatographic steps that had been used in the production of clinical supplies of a fusion protein. The first case is a hydrophobic interaction chromatography step that had been used to separate the mature fusion protein from a prospecies variant that had an additional 41 amino acids of the pro-peptide bound to the product. The process step exploits the greater hydrophobicity of the prospecies to bind both species during the load and then selectively elute the product while leaving the prospecies bound to the resin. In the second case, the wash step of an anion-exchange chromatography column was used to remove charge variants having reduced levels of tyrosine sulfation.

Modeling of Hydrophobic Interaction Chromatography

The ability of the batch-binding model system to separate a product and related impurity was tested. A batch-binding system with 100 μL of hydrophobic-interaction chromatographic (HIC) resin, Phenyl Toyopearl 650C, was compared to a laboratory-scale column that was 1.1 cm $\times$ 13.5 cm, or

12.8 mL, while the manufacturing-scale column was 60 cm × 13.5 cm, or 38 L. These columns were approximately 128-fold and 380,000-fold larger than the resin bed volume used in the filter-plate, respectively.

The lab-scale column was loaded with the Fc-fusion protein containing 6.5% prospecies in 1.2 M ammonium sulfate, 20 mM Tris, pH 7.4. The column was washed with 5 column volumes (CV) of the loading buffer prior to elution with 0.48 M ammonium sulfate, 20 mM Tris, pH 7.4.

In the batch-binding model, the protein mixture, with 6.5% prospecies, was incubated and mixed with resin in a 50 mL conical tube (#352070, Becton Dickinson, Franklin Lakes, NJ) under column loading conditions (1.2 M ammonium sulfate, 20 mM Tris, pH 7.4) for 1 h. Bulk batch binding was performed to minimize well-to-well variability. The 1 h incubation time, though likely unnecessary as shown above, was to assure that the system was near equilibrium. A resin volume of 100 μ L (as a 50% slurry) was distributed to each well of a 25 μ m pore-size filter-plate. After initial distribution, the filter-plate was evacuated in a centrifuge at 180 g. Each well in the filter-plate was then washed in three stages of 300 μ L of the 1.2 M ammonium sulfate load/wash solution. Each stage consisted of buffer addition, shaking at 1300 rpm for 20 min, and evacuation in a centrifuge at 180g. The wells were then eluted in a similar fashion with three stages of 300 μ L of the column elution buffer (0.48 M ammonium sulfate, 20 mM Tris, pH 7.4). The wells were then stripped with three stages of 300 μ L of 20 mM Tris, pH 7.4.

Both the column and the batch-binding model were analyzed by a UV spectrophotometer and by Reversed-Phase HPLC to determine total concentration and prospecies level.

The manufacturing process purified an Fc-fusion protein from its prospecies with HIC resin. The prospecies had a leucine-rich 41-residue polypeptide, which had not been cleaved from the N-terminus. The prospecies therefore bound to the HIC resin more tightly than the product, which is the basis for the separation. Figure 2 depicts a lab-scale chromatogram of the process.

At both lab and production scale, a small amount of impurity did not bind to the column during the load, as seen in Figure 2. No protein was detected in the wash fractions. The elution peak (typically 0.7 column volumes) followed a prepeak of approximately 1CV, and contained the Fc-fusion product largely free of prospecies. The strip contained the majority of the prospecies loaded onto the column. The overall recovery of the Fc-fusion and prospecies averaged 86% for a clinical manufacturing campaign. Typical prospecies levels in the product peak were approximately 0.5%. These results are typical of the performance of this operation in a 60 × 13.5 cm, 38 L column.

The system in the 96-well filter-plate performed similarly, with the exception of the lack of A₂₈₀ absorbing material detected in the supernatant prior to distributing the loaded resin into the filter-plate. The elution buffer of 0.48 M ammonium sulfate was applied directly to the well during the three elution stages, followed by the strip buffer in three successive strip stages (see results in Fig. 3). Due to hold-up of the higher loading salt concentration, the first stage of the

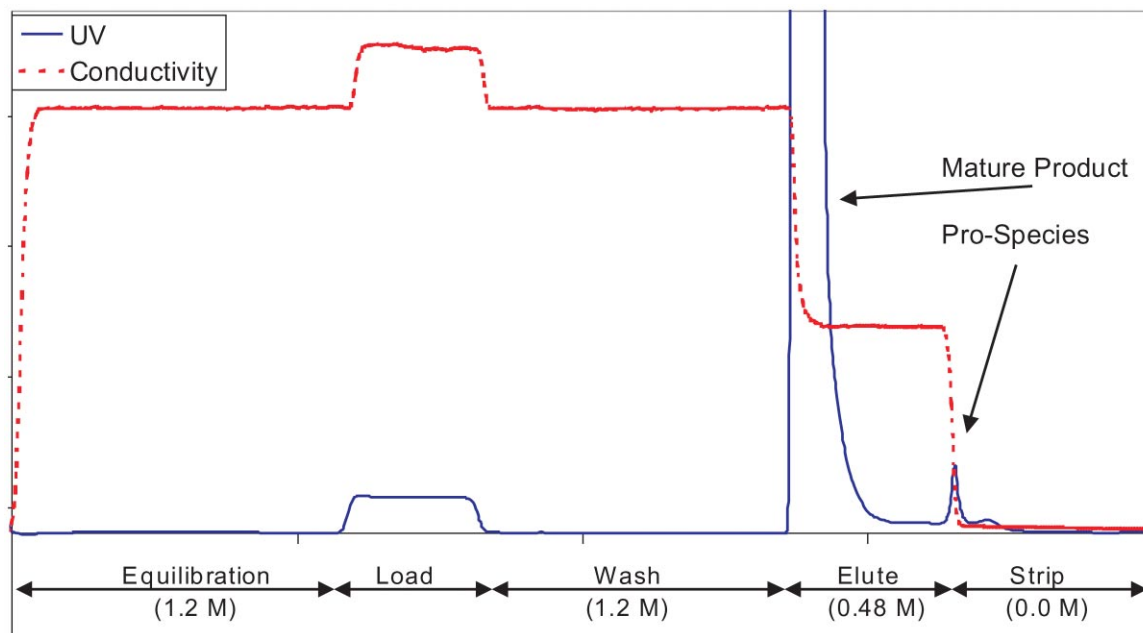


Figure 2. Lab-scale process chromatogram for separation of the Fc-fusion protein from its pro-species on a HIC column. The molarity of ammonium sulfate at each step is noted. [Color figure can be seen in the online version of this article, available at www.interscience.wiley.com.]

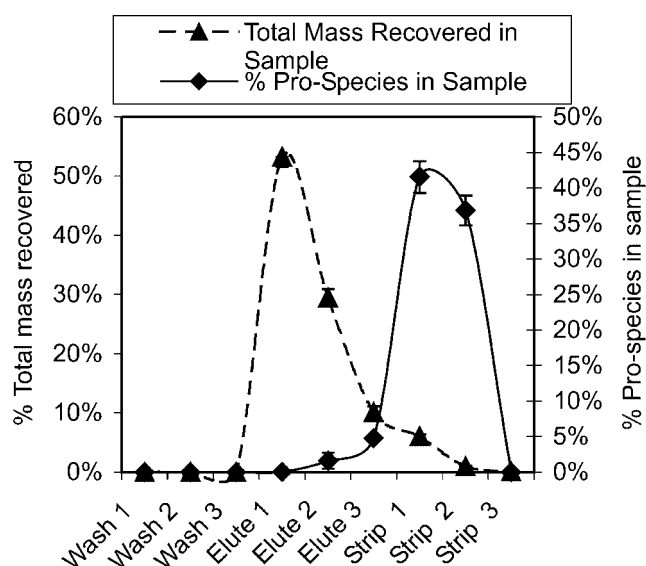


Figure 3. Data from the separation of the Fc-fusion protein from its pro-species on HIC resin in a filter-plate well. The product is eluted as expected with enriched impurity coming off the resin in the strip.

elution had a concentration of 0.61 M ammonium sulfate (see Eq. 2). The second stage was within 3% of the expected elution buffer concentration at 0.50 M ammonium sulfate. Both the second and third elution stages had higher levels of pro-species than seen in the column. The strip of each well had the majority of the loaded pro-species. The total mass recovery was 105%. This recovery is slightly higher than the typical lab-scale and manufacturing-scale runs, but within the range variability seen for this step. The pro-species level of the pooled elution stages was 1.1%, which was slightly higher than the 0.5% observed in the columns runs.

A comparison of the filter-plate purification and the lab-scale and industrial scale column-based purification demonstrated that purification performance could be approximated on a filter-plate with only 100 μ L of resin per well. The filter-plate elution profile and column profile differed primarily due to the quantifiable hold-up volume from stage to stage. The elution volume was larger as a result of this hold-up, both due to the higher salt in the elution stages, and stage-to-stage hold-up of the eluted protein. Secondary effects, such as the increased levels of the pro-species during the third stage of the elution series, were due to the expected and quantifiable effects of the single-stage equilibrium in the filter-plate compared with the multiple equilibrium stages found in chromatographic columns. The qualitative agreement between the HTS data and the column results indicate that batch-binding data obtained in a HTS format can be used to guide chromatography column development.

Based on the batch-binding data, the separation factor (IUPAC Compendium of Chemical Technology, 1997)

between the pro-species and the Fc-fusion protein was determined to be 15 at 0.48 M ammonium sulfate. This separation factor is quite large by industrial separation standards, indicating a relatively easy separation.

Modeling of Anion-Exchange Chromatography

The ability of the filter-plate system to separate a product and related impurities was tested with another well-characterized process that had been used to purify clinical-grade material. This AEX step removed six hyposulfated product isoforms from the fully sulfated product with a 20 column-volume low-pH wash. The product was eluted at neutral pH with high salt. The hyposulfated isoforms are less acidic, and therefore bind less tightly to the AEX resin. The HTS system used the same conditions, solutions, and resin as the large-scale chromatographic system. The performance of the HTS system was compared to that of the large-scale system by evaluating the purification of the product from the six hyposulfated isoforms.

The anion-exchange (AEX) laboratory scale column was 1.1×12 cm, or 12 mL, while the manufacturing-scale column was 80×12 cm, or 60 L, which was approximately 120-fold and 600,000-fold larger, respectively, than the 100 μ L resin volume used in the filter-plate. The column was loaded with the Fc-fusion protein containing 16.5% hyposulfated species in 20 mM Tris, 150 mM NaCl, pH 7.5. The column was washed with 20 column volumes (CV) of 20 mM sodium citrate, 200 mM NaCl, pH 3.6, or pH 3.8, or pH 4.0. The product was then eluted with 1 M NaCl, 50 mM Tris, pH 7.5.

In the batch-binding model for anion-exchange chromatography, the protein mixture, with 16.5% hyposulfated species, was incubated and mixed with the Q Sepharose Fast Flow in a 50-mL conical tube with the load buffer (20 mM Tris, 150 mM NaCl, pH 7.5) for 1 h to ensure more complete binding and to avoid errors from well-to-well variability. A volume of 100 μ L of resin (as a 50% slurry) was distributed to each well of a 25 μ m pore-size filter-plate. Each well was washed with a single stage of 300 μ L of wash buffer (20 mM sodium citrate, 200 mM NaCl, pH 3.6 or 3.8 or 4.0), and then eluted with multiple stages of 300 μ L elution buffer (50 mM Tris, 1 M NaCl, pH 7.5).

Both the column and the batch-binding model were analyzed by a UV spectrophotometer and by anion-exchange HPLC to determine total concentration and hyposulfated species level.

A lab-scale chromatogram of the AEX chromatography step is shown in Figure 4. During the load, very little product or hyposulfated isoforms flowed through the column. During the low-pH wash, a small peak of hyposulfated material exited the column. This leached material washed off the column throughout the 20 column-volume wash. The product was eluted using a neutral pH, high salt buffer. The recovery of the product in this step was typically 90–95% at manufacturing scale. The high-throughput screen used a

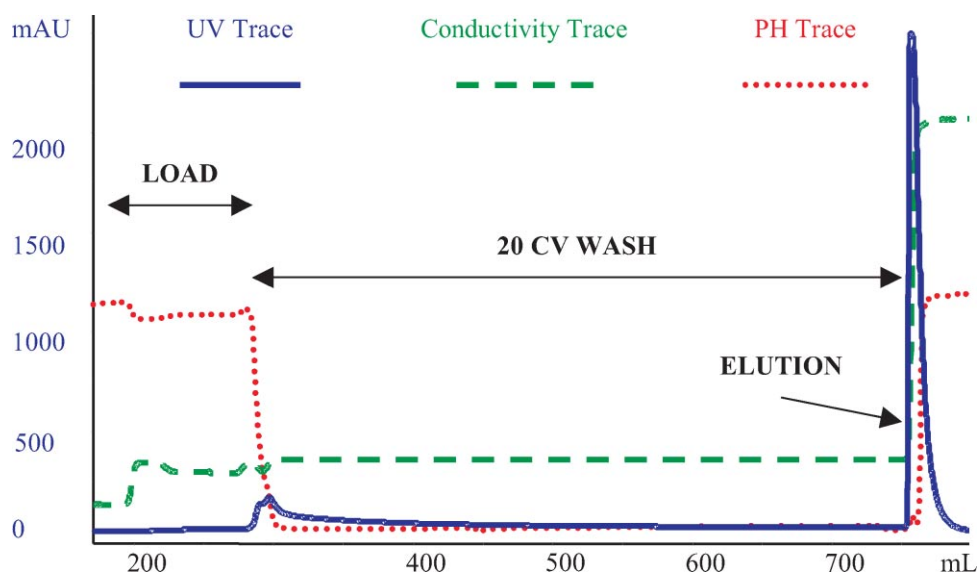


Figure 4. Lab-scale chromatogram of the Fc-fusion protein AEX step demonstrating the removal of hyposulfated species across a 20 CV low salt, low pH, wash and the elution of the fully sulfated product at high salt, neutral pH. [Color figure can be seen in the online version of this article, available at www.interscience.wiley.com.]

single three bed-volume wash, which is significantly smaller and less effective than the 20 column-volume wash used in the column step.

The relative concentrations of product and of hyposulfated species were measured with analytical anion-exchange HPLC. The lab- and manufacturing-scale columns reduced the concentration of hyposulfated species to below 5%. The wash fractions on the HTS showed selective removal of the hyposulfated species, as shown in Figure 5, which reduced the concentration in the product fraction to below 10%. While the impurity level in the HTS product pool was higher than the laboratory- and production-scale columns,

equivalent product purity from a batch-binding system was not expected, as the column process used a continuous 20-column volumes wash with hundreds of theoretical plates while the supernatant volume in a filter-plate stage had only one equilibrium stage and a wash volume of only three resin volumes.

The amount of hyposulfated material removed by the wash depended upon the pH in both the column and in the filter-plate experiments, as shown in Table VI. The amount of hyposulfated material removed by the wash decreased at higher pH levels, due to the increased affinity the protein has for the AEX resin at higher pH. The effectiveness of the wash

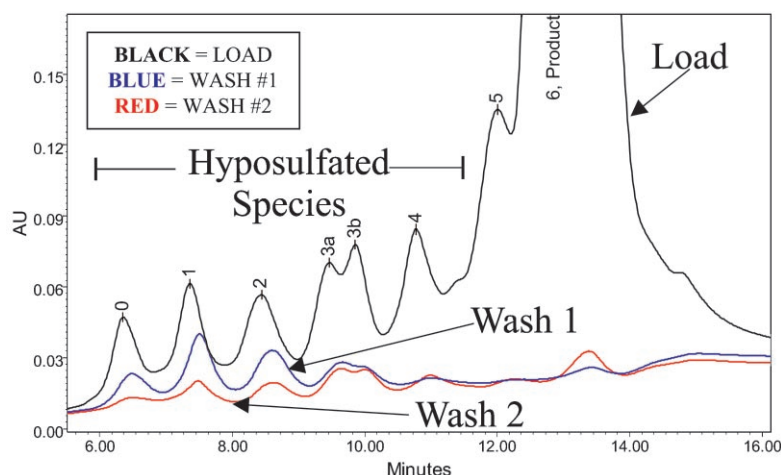


Figure 5. Analytical AEX HPLC chromatograms of the HTS pH 3.8 wash supernatants in the AEX separation with the Fc-fusion protein.

Table VI. A comparison of the level of hyposulfated material in the wash at pH 3.6, pH 3.8, and pH 4 for the Q Sepharose Fast Flow column and the High-Throughput Screen (HTS).

pH	% of hyposulfated species in Wash	
	Column	HTS
3.6	96	49
3.8	94	40
4	84	34

The separation was shown to be sensitive to subtle changes in wash pH. All HTS values are averages of duplicate wells.

in the HTS was significantly less than that in the column due to the lower wash volume (as a fraction of resin volume) used in the HTS and the fact that the well is a single equilibrium stage.

Based on the HTS data, the separation factor (IUPAC Compendium of Chemical Technology, 1997) between the hyposulfated material and the fully sulfated Fc-fusion protein was determined to be over 200 during the wash. While this separation factor is extremely high, the 20 CV wash is required because the hyposulfated material binds to the column tightly during the wash.

Application of High-Throughput Screening Methods to Purification Development

Several applications for development of chromatographic separations have been examined using a high-throughput screening system employing a robotic liquid handler and 96-well filter plates. Almost any solution composition for load, wash, elution and regeneration can be configured by programming the robotic system, and using concentrated stock solutions of excipients which are then mixed in various proportions. These methods are useful for the development of large-scale protein purification unit operations.

Basic thermodynamic information on protein–resin interactions can be obtained from HTS studies, including determination of the equilibrium adsorption isotherm, capacity factors, separation factors, and maximum binding capacity. Because 96 different conditions can be evaluated in a single plate, high-resolution response surfaces can be constructed in a single experiment to provide detailed information for modeling or regression analysis. Comparative evaluations of related resins or proteins are easily accommodated by the 96-well format, allowing head-to-head testing designs that minimize blocking effects or day-to-day variability inherent in lower-throughput systems. The screening of 8 or 12 resins of a single functionality (families of anion-exchange or hydrophobic interaction resins, for instance) is possible using a single plate and an experimental design that surveys several other variables (Kramarczyk et al., 2008; Wensel et al., 2008; Kelley et al., 2008).

Many different modes of chromatography could be adapted to HTS batch binding (later papers in this series will address specific applications to hydrophobic interaction,

hydroxyapatite, and ion-exchange chromatography (Kelley et al., 2008; Kramarczyk et al., 2008; Wensel et al., 2008)), with the exception of size exclusion chromatography, where a successful separation requires hundreds of theoretical plates. The simulation of gradient chromatography is relatively easy, but elution chromatography, employing large step changes in solution composition or excipient concentration, may be difficult. Elution under conditions where there is significant counterligand binding (as in immobilized metal ion affinity chromatography) or pH changes (Protein A chromatography) will be complex, and possibly difficult to translate directly from the HTS system to column operation. Even when accounting for hold-up of the previous solution, and factoring in an overshoot in the first elution stage using Equation (2), the correct composition may not be possible to achieve. Adjusting the composition of the previous cycle may be necessary, for instance by decreasing buffer concentration. This technique has shown utility during evaluation of Protein A chromatographic steps that employ low pH elution (data not shown).

A distinct advantage of the HTS system is the speed with which many different solutions or conditions can be tested. An adsorptive column operation can be modeled by load, wash, and elution cycles in less than 8 h, and a flowthrough step or a set of isotherms can be modeled in only 4 h. These times are nearly two orders of magnitude shorter than a comparable set of column runs. The benefits of the 96-well plate analysis then become apparent, as all individual fractions from the various cycles of the chromatographic operations are already aliquotted into their own plates and are analyzed together.

In addition, the scale-down factor representing the volume of a process scale column in relation to the volume of the resin in the well is much higher than the smallest scale laboratory column. Assuming a lower limit of a 1 cm diameter and 20 cm bed height for a suitably scaled laboratory column (resulting in a column volume of 16 mL), the volume of an individual well is 320-fold smaller than this lab-scale system, and potentially six to seven orders of magnitude smaller than a process column. A single plate with 96 different conditions would still consume only one-third of the material needed for a single laboratory column run. The total resin used during a single 96-run test could be as low as 5 mL. Protein requirements depend upon the methods of detection, as well as desired resin challenge. The protein requirements range from 25 mg (for typical operation near the linear range, assumed to be about 5 mg/mL resin challenge) to 200 mg (representing a 40 mg/mL challenge of a fully loaded adsorption cycle) to as much as 750–1,000 mg (for a non-adsorptive step that challenges the resin to 150–200 mg/mL).

The analytical methods used to assess the supernatant from the wells determine the lowest feasible mass challenge to the resin. The amount of protein needed for a screen has been calculated and is shown in Table VII, assuming that the detection method must detect as little as 5% of the load. Sensitive methods of detection, such as ELISA, could

Table VII. The analytical technique for protein concentration measurement dictates the material required for a high-throughput screen.

Analytical Technique	Detection limit ($\mu\text{g/mL}$)	Mass per plate (mg)
UV-vis absorption	50	30
Bicinchronic acid (BCA) protein binding	20	12
HPLC with UV detection	20	12
Fluorometric Detection Reagent Nano-Orange [®]	10	6
Capillary Electrophoresis	1	0.6
ELISA (for product or impurity)	0.001	0.001

Assuming 5% of the load must be detected, the limit of detection of the assay dictates the protein mass required in each well, and for each plate.

conceivably allow a high-throughput screen with as little as 1 μg of protein. At this level of protein, non-specific interactions between the protein and the microtiter plate may become significant.

Analytical methods also have the potential of being a bottleneck in utilization of HTS in downstream process development. Of the methods shown in Table VII, HPLC and CE are methods with two of the lowest throughput. To fully analyze a 96-well plate for aggregate level using a 20-min HPLC run would require 32 h. Strategies that analyze wells with only protein concentrations in the range of interest as determined by UV absorption can cut this analytical time significantly. The total time of 36 h is, however, manageable for a downstream development lab. Several automated HPLC systems or CE systems could easily be purchased to quickly analyze the samples from a robotic system. This approach is usually not necessary, however, since the amount of data obtained in a 32-h time period via HTS is at least an order of magnitude faster than the development time of the same screens run in column mode.

Despite the many advantages of the HTS system, there are limitations that must be considered during the design of experiments and interpretation of results. The direct translation of the yield and purity data from a simulated chromatographic cycle to a true column operation is not straightforward, as there are several important differences between these systems. The batch-binding in the well represents only a single stage (or theoretical plate), in contrast to a column which will typically have hundreds of plates. Thus, the degree of enrichment observed in the HTS system is likely to be a conservative estimate of the resolving power of a well-packed column. Flow in a packed column will also influence the dynamic capacity of the resin, which will be lower than the equilibrium capacity measured with the HTS batch-binding system. However, the trends in purity and yield obtained by the HTS match those seen in the column, making this technique useful for optimization. HTS may also be useful for robustness analysis of the process within specified control ranges by examining the rate of change of purity or recovery near the final column parameters. Appropriate mass-transfer or kinetic models of the chromatographic separation may be used to convert the information provided by the HTS studies into quantitative predictions of column performance.

Because of these differences between a well and a column, the data obtained from the screen provides only a range of parameters to study in column mode, not necessarily the optimal solution. Starting with the best conditions identified by the HTS, and an understanding of the effect each parameter has on purity and yield, the development scientist can quickly establish optimal column conditions. This optimum will be based not on the few column experiments typical in purification development, but on a hundred or more conditions tested in the high-throughput screen. The extent of screening provides some assurance that a more global optimum has been obtained, and that the solution is robust.

Conclusion

The use of a robotic liquid dispensing systems combined with a miniaturized batch-binding format has the potential to accelerate the development of chromatographic steps for protein purification if appropriate measures are taken to assure accuracy, mix properly, allow for kinetic effects, and account for liquid hold-up from one stage to the next.

Volumes of 50 and 100 μL of eight commercially available chromatographic resins were dispensed with a variability of 4% into a 96-well plate with a standard robotic liquid-handling system. The binding kinetics of an antibody were measured to establish a minimum contact time of 20 min for the batch binding. The volume of liquid remaining after well evacuation via centrifugation was determined to have a significant impact on solute and protein mass balance if neglected. The performance of the HTS compared favorably with that of hydrophobic interaction and anion-exchange lab-scale and production-scale chromatographic columns. Since the HTS utilizes a single stage batch-binding step, the product pools from the HTS had higher impurity levels than those from the column runs. The recovery of product and impurities was found to be quantitative and reproducible, and indicated that the amount of material lost due to non-specific binding was small.

The HTS system allows the rapid investigation chromatographic conditions involving hundreds of combinations of solution and resin combinations with small quantities of material. While the results of the HTS batch binding are not

identical to those seen in chromatographic columns, they do enable the trending of purity and recovery with process parameters. These trends are sufficient to assist in chromatographic step development by providing extensive, multivariate datasets that can be correlated to actual column performance. Process development groups may now supplement traditional column studies used for process definition and optimization with data on yield and purity derived from this high-throughput screening methodology.

Q column data provided by Russell Sphritzer. HIC column data provided by Angie Bragdon. Kinetic data, hold-up volume, and resin variability experiments performed by Michael Shamashkin. Dye reproducibility test performed by David Wensel.

References

- Albala JS, Franke K, McConnell IR, Pak KL, Folta PA, Rubinfeld B, Davies AH, Lennon GG, Clark R. 2000. From genes to proteins: high-throughput expression and purification of the human proteome. *J Cell Biochem* 80(2):187–191.
- Brumbaugh EE, Ackers GK. 1971. Molecular sieve studies of interacting protein systems. VII. Direct optical scanning method for ligand-macromolecule binding studies. *Anal Biochem* 41(2):543–559.
- Chapman T. 2005. Pure but not simple. *Nature* 434:795–798.
- Charlton H, Galarza B, Leriche K, Jones R. 2006. Chromatography process development using 96-well microplate formats. *BioPharm (June Supp)*:20–26, 42.
- Cramer SM, Rege K, Dordick JS. 2001. Feb. 23, 2001. High Throughput Screening of Potential Displacer Molecules. US. patent application 09/791,317.
- Emili AQ, Cagney G. 2000. Large-scale functional analysis using peptide or protein arrays. *Nat Biotechnol* 18(4):393–397.
- Garr CD, Schultz L, Cameron LM. 2004. Purification and analysis of parallel libraries. In: Kyranos JN, editor. *Purification and Analysis of Parallel Libraries*. Amsterdam: Elsevier.
- Graham EE, Fook CF. 1982. Rate of protein absorption and desorption on cellulosic ion exchangers. *AIChE J* 28(2):245–250.
- Hubbich J, Linden T, Knieps E, Thommes J, Kula M-R. 2002. Dynamics of protein uptake within the adsorbent particle during packed bed chromatography. *Biotechnol Bioeng* 80(4):359–368.
- Kelley BD, Switzer M, Bastek P, Kramarczyk JF, Yu T, Coffman JC. 2008. High-throughput screening of chromatographic separations: 4. Ion Exchange. *Biotechnol Bioeng*, in press.
- Kramarczyk JF. 2003. High-Throughput Screening of Chromatographic Resins and Excipients for Optimizing Selectivity. Medford, MA: Tufts University.
- Kramarczyk JF, Kelley BD, Coffman JC. 2008. High-throughput screening of chromatographic separations: II. Hydrophobic Interaction. *Biotechnol Bioeng* 100:707–720.
- Lacki KM, Bergander T, Gronberg A, Oberg K. 2007. High throughput process development: Determination of dynamic binding capacity using microtiter plates filled with chromatography resins. Abstracts of Papers, 234th ACS National Meeting. Boston, MA, United States: August 19–23, 2007:BIOT-470.
- Lesley SA. 2001. High-throughput proteomics: protein expression and purification in the postgenomic world. *Protein Expr Purif* 22(2):159–164.
- Li R, Dowd V, Stewart DJ, Burton SJ, Lowe CR. 1998. Design, synthesis, and application of a protein A mimetic. *Nat Biotechnol* 16(2):190–195.
- Linden T, Junglof A, Kula M-R, Thommes J. 1999. Visualizing two-component protein diffusion in porous adsorbents by confocal scanning laser microscopy. *Biotechnol Bioeng* 65(6):622–630.
- Liu J, Yang T, Ladiwala A, Cramer SM, Breneman CM. 2006. High throughput determination and QSER modeling of displacer DC 50 values for ion exchange systems. *Sep Sci Technol* 41(14):3079–3107.
- Lowe CR. 2001. Combinatorial approaches to affinity chromatography. *Curr Opin Chem Biol* 5(3):248–256.
- Mazza CB, Rege K, Breneman CM, Sukumar N, Dordick JS, Cramer SM. 2002a. High-throughput screening and quantitative structure-efficacy relationship models of potential displacer molecules for ion-exchange systems. *Biotechnol Bioeng* 80(1):60–72.
- Mazza CB, Rege K, Breneman CM, Sukumar N, Dordick JS, Cramer SM. 2002b. High-throughput screening and quantitative structure-efficacy relationship models of potential displacer molecules for ion-exchange systems. *Biotechnol Bioeng* 80(1):60–72.
- Merson JR. Pfizer Lt, assignee. 1991. An Assay Tray Assembly. patent application 91303083.9.
- Pottorf RS, Player MR. 2004. Process technologies for purity enhancement of large discovery libraries. *Curr Opin Drug Discov Devel* 7(6):777–783.
- Rege K, Tugcu N, Cramer SM. 2003. Predicting column performance in displacement chromatography from high throughput screening batch experiments. *Sep Sci Technol* 38(7):1499–1517.
- Rege K, Ladiwala A, Cramer SM. 2005a. Multidimensional high-throughput screening of displacers. *Anal Chem* 77:6818–6827.
- Rege K, Pepsin M, Flacon B, Steele L, Heng M. 2005b. High-throughput process development for recombinant protein purification. *Biotechnol Bioeng* 93(4):618–630.
- Saraswat LD, Zhang H, Hardy LW, Jones SS, Bhikhabhai R, Brink C, Bergenstrahle A, Haglund R, Gallion SL. 2005. Affinity ligand selection from a library of small molecules: assay development, screening, and application. *Biotechnol Prog* 21(1):300–308.
- Scopes R. 1987. *Protein Purification: Principles and Practice*. New York: Springer.
- Shepard CC, Tiselius A. 1949. The chromatography of proteins. The effect of salt concentration and pH on the adsorption of proteins to silica gels. *Discuss. Faraday Soc.* 7:275–285.
- Stein A, Kiesewetter A. 2006. Cation exchange chromatography in antibody purification: pH optimized screening for optimized binding and HCP removal. *J. Chromatogr. B* 848:151–158.
- Sunasara KM, Xia F, Gronke RS, Cramer SM. 2003. Application of hydrophobic interaction displacement chromatography for an industrial protein purification. *Biotechnol Bioeng* 82(3):330–339.
- Tecan. 2008. RSP Operating Manual Report nr 390 783 V2.
- Thiemann J, Jankowski J, Rykl J, Kurzwski S, Pohl T, Wittmann-Liebold B, Schluter H. 2004. Principle and application of the protein-parameter screening system. *J. Chromatogr. A* 1043:73–80.
- Venn RF, Merson JR, Cole S, Macrae P. 2004. 96-well solid-phase extraction: A brief history of its development. *J. Chromatogr. B* 817:77–80.
- Wenger MD, DePhillips P, Price CE, Bracewell DG. 2007. An automated microscale chromatographic purification of virus-like particles as a strategy for process development. *Biotechnol Appl Biochem* 47:131–139.
- Wensel DL, Kelley BD, Coffman JC. 2008. High-throughput screening of chromatographic separations: 3. Monoclonal Antibodies on Hydroxyapatite. *Biotechnol Bioeng*, in press.
- Wierling PS, Bogumil R, Knieps-Grünhagen E, Hubbich J. 2007. High-throughput screening of packed-bed chromatography coupled with SELDI-TOF MS analysis: Monoclonal antibodies versus host cell protein. *Biotechnol Bioeng* 98(2):440–450.
- Yoshida H, Yoshikawa M, Kataoka T. 1994. Parallel transport of BSA by surface and pore diffusion in strongly basic chitosan. *AIChE J* 40(12):2034–2044.